

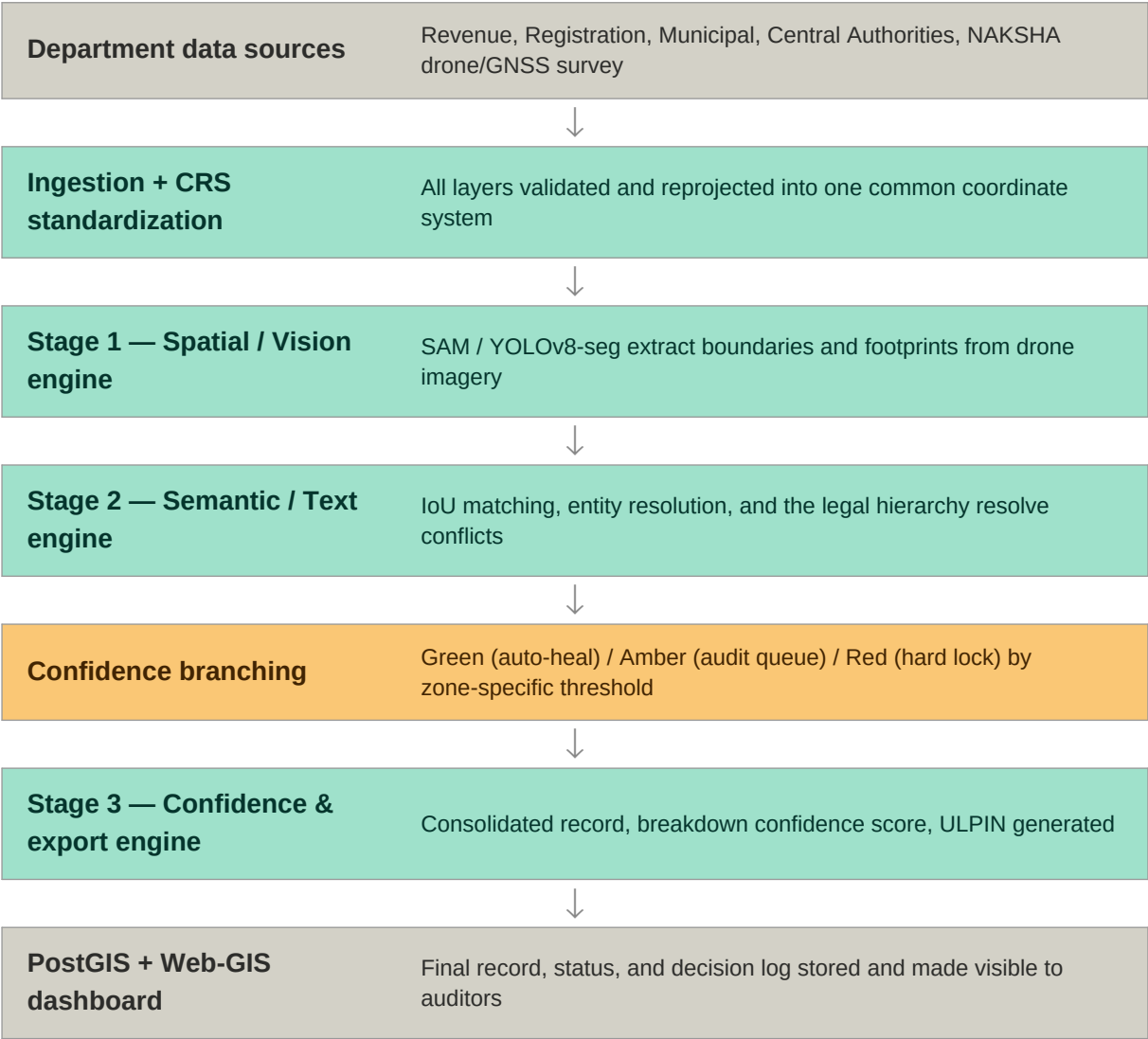
Automated Integration and Intelligent Harmonization of Multi-source Geospatial Data for Urban Land Record Management

Pipeline walkthrough — Smart India Hackathon 2026, Problem Statement 2013

1. What this pipeline does

Urban land records are split across five or more government departments — Revenue, Stamp & Registration, Urban Local Bodies, Central Authorities, and the NAKSHA drone survey programme — each keeping its own version of the truth in its own format. This platform automatically ingests all of these sources for a given parcel, reconciles their disagreements using a fixed legal hierarchy and a measurable distance threshold, and produces one trustworthy, status-tagged record per parcel — without ever silently overwriting a citizen's legal boundary.

2. The pipeline at a glance



3. Where the data actually comes from

Before any processing happens, real files/feeds are pulled from each department — this is what "multi-source" means in practice: five separate hand-offs, five separate formats, arriving via file export or department API/scrape.

Department / agency	What it provides	Typical format
Revenue Department (Bhulekh / Bhu-Naksha)	Legacy cadastral vector, Khasra number, owner name	Digitized shapefile / legacy DB
Stamp & Registration (IGRSUP)	Latest sale deed / ownership transaction	Structured record / PDF
Municipal Corporation / ULB	Property tax ID, building footprints, zoning layer	Municipal GIS shapefile
NAKSHA drone survey agency	Orthomosaic (ORI), DSM/DTM	GeoTIFF
GNSS / CORS field survey	High-precision ground-truthed corner coordinates	Survey-grade point data

4. Worked example — tracing one parcel through the pipeline

To make this concrete, here is a single parcel — Khasra No. 245/2, a suburban residential plot — followed through every stage.

Step 1 — Ingestion + CRS standardization

All five source layers for this parcel (Revenue vector, Registration record, Municipal footprint/zoning, drone ORI, GNSS points) are validated for geometry errors and reprojected into one common coordinate system. Without this, a boundary point from the old Bhu-Naksha map and a point from the drone survey cannot be compared in real meters.

Step 2 — Stage 1: Vision engine

A segmentation model (SAM / YOLOv8-seg) runs over the drone orthomosaic and draws a fresh boundary polygon for this parcel based purely on what is physically visible on the ground. This new shape is tagged Priority 2 under the legal hierarchy.

Step 3 — Stage 2: Semantic engine

The engine now holds up to three competing shapes for the same parcel: GNSS/CORS ground-truthed corners (Priority 1), the AI-extracted boundary (Priority 2), and the legacy Bhu-Naksha shape (Priority 3). IoU matching confirms all three refer to the same real-world parcel, then Priority 2 and Priority 3 are each snapped independently onto the immutable Priority 1 anchor. Text attributes (owner name variants, Hectares↔Sq. Yards) are reconciled in parallel.

Step 4 — Confidence branching

The measured discrepancy between sources for this parcel comes out to 18 cm. Because this parcel is zoned residential, its threshold band is ≤ 15 cm for Green and up to 30 cm for Amber. An 18 cm gap falls in the Amber band: the boundary is tentatively merged so operations are not blocked, but a system flag routes it to an offline audit queue rather than finalizing it silently.

Step 5 — Stage 3: Confidence & export engine

A consolidated record is built: merged geometry, reconciled attributes, and a breakdown confidence score — for example Boundary Accuracy 82% (docked for the Amber flag) and Ownership Certainty 100% (Registration and Revenue names matched cleanly). Because the parcel is Amber and not Green, its ULPIN is generated but marked pending audit rather than fully finalized.

Step 6 — Storage and final output

The final geometry, attributes, confidence breakdown, and status (AUDIT_PENDING) are written as one row in the PostGIS parcels table, alongside a decision log entry recording the parcel ID, threshold used, distance measured, and branch taken. The Web-GIS dashboard, reading from PostGIS, displays this parcel color-coded amber so a human auditor can review the flagged discrepancy and either approve it (status becomes VERIFIED) or escalate it (status becomes LOCKED_DISPUTE).

In short: five siloed department datasets go in, and one trustworthy, status-tagged, auditable parcel record comes out — stored centrally in PostGIS and visible on the dashboard to whichever human needs to act on it next.